Project : Digital Farming Assistant

Report Title : Class-based Modelling and Product Backlog of Digital Farming Assistant

Submitted By

Group Name : **CodeX**
Group Number : **2**

2023-2-60-098	Kazi Mazharul Islam Mridul
2023-3-60-488	MD. Mujahidul Islam
2023-3-60-493	Samiha Ayman Soveridgn
2023-3-60-497	Utsab Roy

List Of Nouns In Digital Farming Assistant

SL No.	Noun
1.	User
2.	Farmer
3.	Farmer ID
4.	Farmer Name
5.	Language Preference
6.	Offline Accessibility
7.	Location ID
8.	Agronomist
9.	Agronomist ID
10	Expertise Area
11.	Registration Date
12.	Crop ID
13.	Crop Name
14.	Image
15.	Upload Status
16.	Image ID
17.	File Path
18.	Forecast Type
19.	Weather Data
20.	Rain
21.	Wind Speed
22.	Temperature
23.	Weather Alert
24.	Location
25.	Crop
26.	Humidity
27.	Forecast Date
28.	Crop Profile
29.	Planning Tips
30.	Pesticide ID
31.	Soil Type
32.	Planting Method
33.	Admin

34.	Admin ID
35.	Role
36.	Soil Requirement
37.	Diagnosis
38.	Diagnosis ID
39.	Result
40.	Figure

41.	Fertilizer
42.	Pesticide
43.	Post
44.	Comment
45.	Like
46.	Notification
47.	Message
48.	Email
49.	Phone Number
50.	Password
51.	ID
52.	Authentication
53.	Login
54.	Auth Method
55.	Logout
56.	Translation service
57.	Weather API
58.	AI Assistant
59.	Chatbot
60.	Model Type
61.	Email/SMS
62.	Image Analysis Service
63.	Disease
64.	Disease ID
65.	Treatment Options
66.	Crop Condition
67.	Weather Forecast
68.	Limitation
69.	Irrigation
70.	Internet

71.	Community Forum
72.	Forum Post ID
73.	Experience
74.	Offline Access
75.	Keyword Search
76.	Database
77.	Language
78.	AI/ML Model
79.	Schedule irrigation
80.	Smartphone
81.	Harvesting
82.	Rot percentage
83.	Crop library
84.	Guidance
85.	User Interface
86.	system update
87.	Crop tips
88.	Pest outbreak
89.	Field Layout
90.	Cultivation
91.	Registration
92.	Profile
93.	forgotten password
94.	Name
95.	recommendation
96.	Timestamp
97.	Yield Estimate
98.	Symptoms
99.	severity
100.	description

101.	accuracy
102.	Alert type
103.	Request
104.	response
105.	tools
106.	Keyword Search
107.	Crop List

List Of Verbs In Digital Farming Assistant

SL No.	Verb
1.	Create Account
2.	Register
3.	Provide Information
4.	Provide User type
5.	Authenticate
6.	Login
7.	Confirm Signup
8.	Logout
9.	Capture Image
10.	Take Photo
11.	Upload
12.	Analyze
13.	Queue
14.	Detect
15.	Diagnose
16.	Choose
17.	Click
18	Decide
19.	Estimate
20	Explore
21	Harvest
22	Handle
23	Guide
24	Generate
25	Recommend
26	Verify
27	Comment
28	Post
29	Like
30	Search
31	Filter
32	View

33	Save
34	Track
35	Notify
36	Send
37	Receive
38	Schedule
39	Forecast
40	Advice
41	Manage
42	Update
43	Reset
44	Store
45	Link
46	Display
47	Provide
48	Enable
49	Confirm
50	Access
51	Attach
52	Reply
53	Feedback
54	Plan
55	Record
56	Reduce
57	Track
58	View
59	Predict
60	Irrigate

Class Cards :

Class : Farmer	
Attributes	**Methods:**
- farmerID + name + phoneNumber - languagePreference - registrationDate - profileImage - locationID - offlineAccessEnabled - accountID	+ registerProfile() + uploadImage() + viewCropHistory() + postToForum() + saveCropProfile() + toggleOfflineAccess() + updateLocation() + chatbot()
Responsibility	**Collaborator**
• Manage personal profile and authentication • Upload crop images for diagnosis • View cultivation guidance and crop history • Participate in community forum • Receive weather alerts and notifications • Access crop library and profiles	• Authentication • Image • Diagnosis • Cultivation • Community Forum • Notification • Crop Profile • Chat Bot

Class : Agronomist	
Attributes	**Methods:**
- agronomistID + name + phoneNumber + expertiseArea - verifiedStatus - profileImage - location - languagePreference -verificationDate	+ verifyDiagnosis() + addForumComment() + createCropProfile() + moderateForum() + updateExpertise() + viewFarmerCultivation()
Responsibility	**Collaborator**
• Verify and validate diagnoses • Provide expert recommendations • Advice in community forum • Share expertise in specific areas • View farmer cultivation data • Upload educational content	• Diagnosis • Community forum • Crop • Profile • Notification • Authentication

Class : Admin	
Attributes	**Methods:**
- adminID + name + email + phoneNumber - expertiseArea - role - lastLogin - location - ProfileImage	+ manageUser() + moderateForum() + updateSystemConfig() + viewSystemLogs() + generateAnalytics report() + manageAuthenticationSettings()
Responsibility	**Collaborator**
• Manage System Users • Monitor and moderate Forum activities • Update System Configurations • View System logs and analytics • Manage authentication settings • Handle escalated issues ·	• Authentication • Community Forum • Farmer • Agronomist • SystemLog

Class : Crop	
Attributes	**Methods:**
- cropID + name + season + soilType + growthDuration - commonDiseases - recommendedFertilizers - recommendedPesticides + tips - imageReference	+ getCropDetails() + findRelatedDiseases() + getGrowingSeason() + SuggestCultivationMethods() + filterBykeyword()
Responsibility	**Collaborator**
• Store basic crop information • Link to multiple crop profiles • Reference common diseases • Provide cultivation requirements • Support library search functionality	• Crop Profile • Disease • Cultivation • Image

Class : Crop Profile	
Attributes	**Methods:**
-profileID - cropID + plantingMethod + soilRequirement + fertilizerRecommendation + pesticideRecommendation + fieldLayoutGuide + OfflineSavedFlag - lastViewedDate - createdBy - createdDate	+ saveForOffline() + updateProfile() + getPlantingGuidance() + getFieldLayout() + trackUsage()
Responsibility	**Collaborator**
• Store detailed cultivation information • Provide planting and care guidance • Link specific field layouts • Support offline saving • Track usage and preferences	• Crop • Farmer • Agronomist • Cultivation

Class : Image	
Attributes	**Methods:**
- imageID - farmerID + timestamp - filePath + cropType + symtopsDetected + fieldLocation - OfflineFile - cropID	+ uploadToServer() + queueForOfflineUpload() + analyzeImage() + getImageMetadata() + updateStatus()
Responsibility	**Collaborator**
• Store and manage crop images • Track upload status • Link to farmer and crop • Provide metadata for analysis • Manage File storage paths	• Farmer • Diagnosis • Crop • AI Assistant

Class : Disease	
Attributes	**Methods:**
- diseaseID + name + symptoms + severityLevel + affectedCrops + treatmentOptions - pestOutbreakFlag - rotPercentageRange - detectionAccuracy + preventionTips	+ getSymptoms() + SuggestTreatments() + checkAffectedCrops() + updateOutbreakStatus() + getPreventionGuidelines()
Responsibility	**Collaborator**
• Store disease characteristics • Provide symptom descriptions • List treatment options • Track affected crops • Provide prevention tips	• Diagnosis • Crop • AI Assistant

Class : Weather Data	
Attributes	**Methods:**
- weatherDataID + location + date + temperature + humidity + rainfall + windSpeed + forecastType - source - timestamp - alertTriggered	+ fetchForecast() + checkForAlerts() + getHistoricalData() + updateWeatherData() + calculateGrowingConditions()
Responsibility	**Collaborator**
• Store current and forecast weather • Trigger weather alerts • Provide location-specific data • Track data sources and timestamps • Support cultivation planning	• Notification • Cultivation

Class : Notification	
Attributes	**Methods:**
- notificationID - recipientID + type + messageContent - timestamp - readStatus + deliveryMethod - priority - relatedEntityID	+ sendNotification() + markAsRead() + resendFailed() + filterByType() + getUnreadCount()
Responsibility	**Collaborator**
• Deliver alerts and messages • Track delivery status • Support multiple delivery methods • Manage recipient lists • Store message content and timing	• Farmer • Agronomist • Admin • Weather Data • Authentication

Class : Authentication	
Attributes	**Methods:**
- authID - farmerID - adminID - AgronomistID + phoneNumber + authMethod - lastLoginTimestamp - deviceInfo + languagePreference - forgotPasswordFlag - accountStatus	+ login() + logout() + resetPassword() + resendFailed() + updateAuthMethod() + getUnreadCount() + lockAccount()
Responsibility	**Collaborator**
• Responsibilities • Manage user login/logout • Handle password resets • Store authentication methods • Track login history • Manage session tokens • Support multiple auth types	• Farmer • Agronomist • Admin • Notification

Class : AI Assistant	
Attributes	**Methods**
- assistantID - toolUsed + AccuracyScore + responseContent + serviceType - requestID - linkedDiagnosisID - modelType - analysisTimestamp	+ analyzeImage(imageID) + provideGuidance(topic, context) + calculateConfidence(metrics) + updateModel(newModel) + logRequestResponse(request, response) + getRecommendations(context)

Responsibility	Collaborator
• Analyze images for diseases • Provide cultivation guidance • Generate recommendations • Track confidence scores • Manage analysis tools/models • Link to service requests	• Image • Diagnosis • Cultivation • Crop Profile • Weather Data

Class : Community Forum	
Attributes	**Methods:**
- forumID + name + description + category -moderatorIDs +totalPosts +totalComments +searchableFlag -creationDate -lastActivity	+ createPost(authorID, content) + addComment(postID, authorID, content) + likePost(postID, userID) + searchPosts(keyword) + moderateContent(postID, action) + getForumStats()
Responsibility	**Collaborator**
• Manage discussion posts • Handle comments and likes • Support search functionality • Moderate content • Track user engagement • Categorize topics	• Farmer • Agronomist • Admin

Class : Cultivation	
Attributes	**Methods:**
-cultivationID -farmerID - cropID +startDate +endDate +fieldLocation +irrigationSchedule +harvestingDate +soilType - yieldEstimate - actualYield - status	+ scheduleIrrigation(dates, amounts) + updateGrowthStage(stage) + recordHarvest(yield, date) + calculateYieldEstimate() + getWeatherImpact() + adjustSchedule(weatherData)
Responsibility	**Collaborator**
• Track farming activities • Schedule irrigation • Monitor growth stages • Estimate yields • Record harvest data • Link to weather conditions	• Farmer • Crop • Weather Data • AI Assistant • Crop Profile

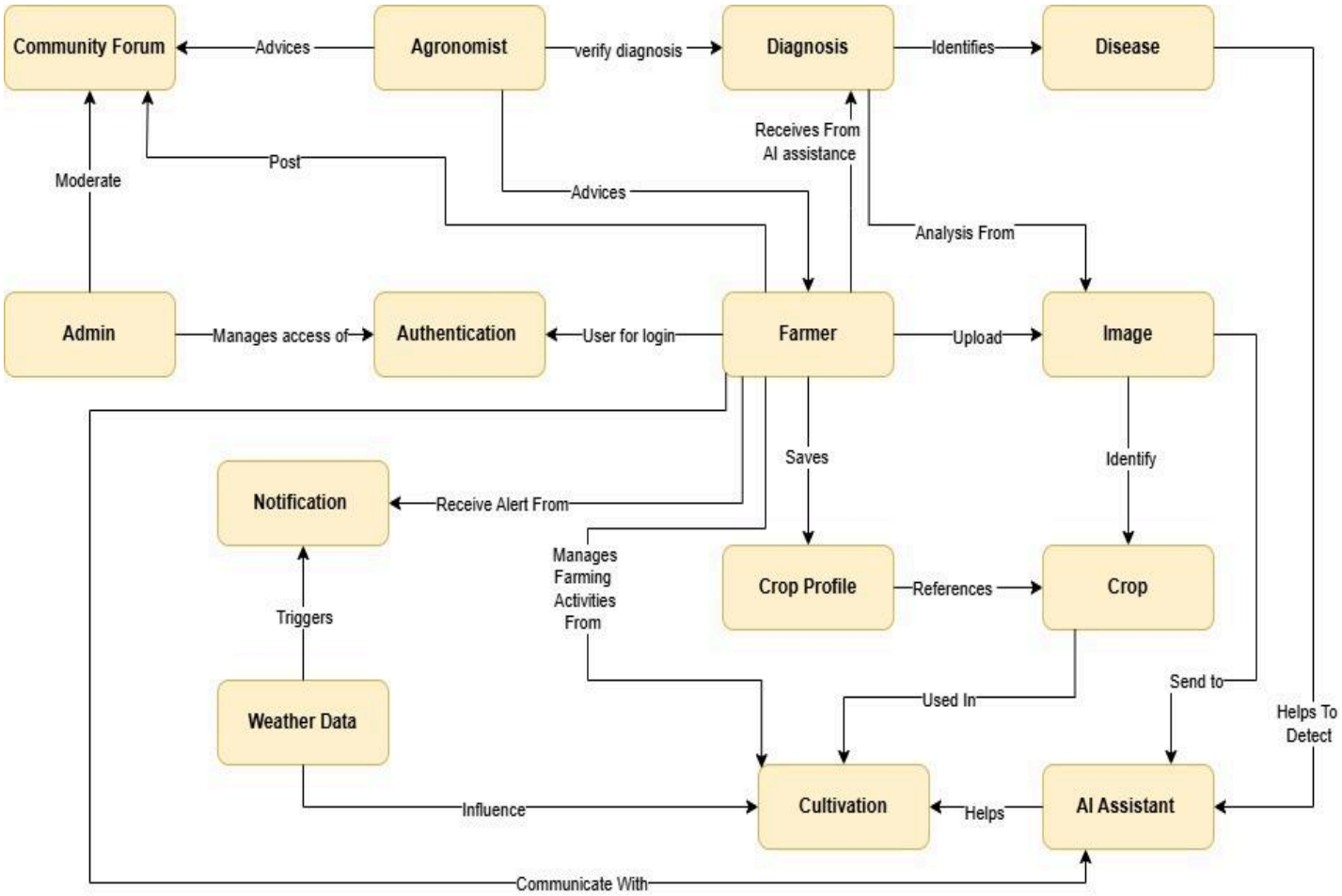


Figure : CRC Diagram for all classes

Link: https://drive.google.com/file/d/1SoxOe6U5O7eyWJQbBZGbHKJK5ER6r704/view?usp=sharing